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DEVICE FOR COOLING A LIVEWELL

Abstract

A modular apparatus, system, and method for managing the temperature of a livewell or other container housing water, and an associated method for installing the same. The system featuring a self-contained fluid line in communication with a first fluid line, which is in fluid communication with the livewell. The modular apparatus generally featuring the system and an enclosure. The enclosure can be located adjacent to the livewell and detachably coupled to a fishing vessel. The modular apparatus and system can further comprise a condenser, which can be a forced air coil unit, a liquid-cooled unit, or a cold plate heat sink unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. patent application Ser. No. 17/816,845, filed Aug. 2, 2022, entitled “Device for Cooling a Livewell,” which is continuation-in-part application of U.S. application Ser. No. 16/723,084, filed Dec. 20, 2019, entitled “Device for Cooling a Livewell,” which claims priority to U.S. Provisional Patent Application No. 62/782,863, filed Dec. 20, 2018, entitled “Device for Cooling a Live Well of a Boat”. The entire disclosure, including the specification and drawings, of the above-referenced applications are incorporated herein by reference.

FIELD

[0002] The present invention relates generally to fishing equipment and, more particularly, to an apparatus and system, and associated method for installing the same, for managing the temperature of a livewell or other container housing water.

BACKGROUND

[0003] Conventional livewell designs lack adequate means for effectively managing the temperature of the various liquids contained therein and fail to adequately and effectively preserve the health of fish and other aquatic animals that may be temporarily stored in the livewell. Although there are some means for managing the temperature of water contained within livewells and other similar environments, such designs are large, expensive, and incapable of managing the temperature with the level of precision required to preserve certain fish, including sport or game fish, live bait, and other aquatic animals.

[0004] Existing means for managing the temperature of water contained within livewells and other similar environments include the use of ice, chemicals, and thermoelectric means, which can be cost-prohibitive and not a practical option for fishers, whether professional or amateur. Further, such known means do not provide means for readily scaling their use to larger or smaller applications, as needed. Nonetheless, the temperature of water contained within livewells and other similar environments must be regulated for various reasons. For example, it may be necessary to regulate the temperature to extend the life of caught fish and other aquatic animals. Alternatively, it may be necessary to regulate the temperature to accommodate various the environmental demands required for certain species of fish and other aquatic animals, such as fish, crayfish, shrimp, and so on. Further yet, it may be necessary to regulate the temperature to provide an optimal environment for fish and other aquatic animals to preserve the health of the fish and other aquatic animals in temporary and mobile storage.

[0005] Accordingly, a need exists for an improved apparatus and associated method for managing the temperature of a livewell provided for preserving the health of fish and other aquatic animals in temporary and mobile storage, including, without limitation, in a manner that is more efficient, cost-effective, adaptable, and compact while also rendering a higher level of precision over known means.

SUMMARY OF THE INVENTION

[0006] Disclosed herein is a modular apparatus and system for managing the temperature of a livewell or other container housing water, and an associated method for installing the same. The modular apparatus can generally comprise a self-contained fluid line, a first fluid line in communication with the self-contained fluid line and configured to provide fluid communication between the modular apparatus and the livewell, a condenser in communication with the self-contained fluid line, and a controller. In some embodiments, the self-contained fluid line can comprise an expansion valve in fluid communication with a compressor, the controller can be configured to communicate with a temperature probe that detects a temperature of water in the livewell and control at least one of the condenser, compressor, and expansion valve to cause the

temperature of the water in the livewell to reach a desired temperature.
[0007] In exemplary embodiments, a method can comprise regulating the temperature of the livewell using temperature regulation components of a modular apparatus, receiving a first signal from a sensor, determining whether the first signal corresponds to one of a plurality of errors statuses, deactivating the temperature regulation components in response to determining that the first signal corresponds to one of the plurality of error statuses, wherein the temperature regulation components include at least one of a compressor, a condenser, and an evaporator, determining whether a second signal indicates that the one of the plurality of error statuses has cleared, and reactivating the temperature regulation components in response to determining that the second signal indicates that the one of the plurality of error statutes has cleared.
[0008] A system can comprise a self-contained fluid line, a first fluid line in communication with the self-contained fluid line and configured to provide fluid communication between the modular apparatus and the livewell, a condenser in communication with the self-contained fluid line, and a controller. In some embodiments, the self-contained fluid line can comprise an expansion valve in fluid communication with a compressor, the controller can be configured to communicate with a temperature probe that detects a temperature of water in the livewell and control at least one of the condenser, compressor, and expansion valve to cause the temperature of the water in the livewell to reach a desired temperature. The system can further include a fishing vessel comprising: the livewell, a second fluid line, an entry port on a fishing vessel, wherein the entry port allows water from a body of water to enter a second fluid line, an exit port connected to the second fluid line that expels water from the second fluid line out into the body of water.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0009] In the accompanying drawings, which form a part of the specification and are to be read in conjunction therewith:
[0010] FIG. **1** is a schematic view of a system including an apparatus for managing the temperature of a livewell in accordance with one embodiment of the present invention;
[0011] FIGS. **2**A and B are each a schematic view of an apparatus for managing the temperature of a livewell in accordance with another embodiment of the present invention;
[0012] FIG. **3** is a schematic view of an apparatus for managing the temperature of a livewell in accordance with yet another embodiment of the present invention;
[0013] FIG. **4** is a schematic view of an apparatus for managing the temperature of a livewell in accordance with even yet another embodiment of the present invention;
[0014] FIG. **5** is a perspective view of an apparatus for managing the temperature of a livewell in accordance with one embodiment of the present invention;
[0015] FIG. **6** is a front view of the apparatus for managing the temperature of a livewell of FIG. **5**;
[0016] FIG. **7** is a first side view of the apparatus for managing the temperature of a livewell of FIGS. **5** and **6**;
[0017] FIG. **8** is a second side view of the apparatus for managing the temperature of a livewell of FIGS. **5-7**;
[0018] FIG. **9** is a second rear view of the apparatus for managing the temperature of a livewell of FIGS. **5-8**;
[0019] FIG. **10** is a top view of the apparatus for managing the temperature of a livewell of FIGS. **5-9**;
[0020] FIG. **11** is a partial perspective view of the interior space of the apparatus for managing the temperature of a livewell of FIGS. **5-10**;
[0021] FIG. **12** is a front view of an apparatus, without an enclosure, for managing the temperature

of a livewell in accordance with one embodiment of the present invention;
[0022] FIG. **13** is a schematic illustration of an example system for managing the temperature of a livewell in accordance with one embodiment of the present invention;
[0023] FIG. **14** is a schematic illustration of a portion of the system of FIG. **13**;
[0024] FIG. **15** is a schematic illustration of an example system for managing the temperature of a livewell in accordance with another embodiment of the present invention;
[0025] FIG. **16** is a schematic illustration of an example system for managing the temperature of a livewell in accordance with yet another embodiment of the present invention;
[0026] FIG. **17** is a diagram depicting an example method for installing a system for managing the temperature of a livewell in accordance with yet another embodiment of the present invention;
[0027] FIG. **18** is top view schematic representation of a fishing vessel containing the system of FIG. **17**;
[0028] FIG. **19** illustrates a method for detecting statuses and controlling the modular apparatus based on detected statutes in accordance with yet another embodiment of the present invention.
DESCRIPTION
[0029] Various embodiments of the present invention are described and shown in the accompanying materials, descriptions, instructions, and drawings. For purposes of clarity in illustrating the characteristics of the present invention, proportional relationships of the elements have not necessarily been maintained in the drawings. It will be appreciated that any dimensions included in the drawings are simply provided as examples and dimensions other than those provided therein are also within the scope of the invention.
[0030] The description of the invention references specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The present invention is defined by the appended claims and the description is, therefore, not to be taken in a limiting sense and shall not limit the scope of equivalents to which such claims are entitled.
[0031] FIG. **1** illustrates an example system **1** that includes a modular apparatus **100**. According to an exemplary embodiment, the system **1** can comprise a fishing vessel **102** or other vehicle that includes a livewell **111**; however, the present invention is not limited to use in a fishing vessel. The modular apparatus **100** can be included within or added to the fishing vessel **102** or added to large land livewells. In an embodiment where the modular apparatus **100** is added to the fishing vessel **102**, the fishing vessel **102** can be retrofitted by disconnecting a first fluid line **120** and a second fluid line **242** and subsequently connecting the first fluid line **120** and the second fluid line **242** to the modular apparatus **100**, and such a method is described in further detail below with regard to FIG. **17**.
[0032] The modular apparatus **100** can couple to a first pump **122** and a second pump **260**. In one embodiment, the first and second pumps **122**, **26** may each be an external pump, such as a pump included as part of the fishing vessel **102**, a manual pump, or an automatic but unconnected pump (e.g., battery powered). In the fishing vessel embodiment, those of skill in the art will recognize that many professional or commercial fishing vessels include or accommodate one or more pumps for pumping lake or ocean water into a livewell or for other fishing or boat-related purposes (e.g., bilge pump). The exemplary embodiments can retrofit the pump and livewell system as shown in FIG. **1**. In some embodiments, a second pump may be omitted as the first pump **122** can provide pumping force for both the first fluid line **120** and the second fluid line **260**.
[0033] As shown, the second pump **260** can be in fluid communication with an entry port **104** via the second fluid line **242**. The entry port **104** can exist on a hull of the fishing vessel **102**, and the second pump **260** can pull in water through the entry port **104** through the second fluid line **242**. The second pump **260** can comprise a pump included as part of the fishing vessel's livewell system or any other pump. While two pumps are shown in FIG. **1**, each end user of the modular apparatus

100 can choose to use more than two pumps for their applications. The fishing vessel **102** can further include an exit port **108**. The exit port **108** can exist on the hull of the fishing vessel **102**, water can be expelled back into a body of water through the exit port **108** via the second fluid line **242**. As explained above, the second pump **260**, the entry port **104**, the second fluid line **242**, and the exit port **108** may be preinstalled within a fishing vessel or may come with the purchase of a particular fishing vessel. However, in another embodiment, the second pump **260**, the entry port **104**, the second fluid line **242**, and the exit port **108** may be aftermarket installed within any boat or water vehicle as well as in applications where the livewell exists on land. In some embodiments, the second fluid line **242** is modular, as a portion of the second fluid line **242** exists within the modular apparatus **100**, and only some of the second fluid line **242** is preinstalled with the fishing vessel **102**.

[0034] The second fluid line **242** can further connect to and extend through the modular apparatus **100**. The second pump **260** can pull water in from the body of water via the second fluid line **242**, the water in the second fluid line **242** can communicate with a condenser **130** within the modular apparatus **100** to receive heat extracted by the modular apparatus **100**, and return heated water back to the body of water via the second fluid line **242**.

[0035] The fishing vessel **102** can further include a livewell **111**. As shown, the first fluid line **120** can couple to the livewell **111** and carry water exiting the livewell **111** to the modular apparatus **100** and transmit chilled fluid to the livewell **111**. The first fluid line **120** can further couple to a first pump **122** to circulate water between the modular apparatus **100** and the livewell **111**, but the first pump **122** may be omitted in embodiments where a single pump moves fluid within both the first fluid line **120** and the second fluid line **242**. By circulating fluid between the livewell **111** and the modular apparatus **100**, a water temperature within the livewell **111** may be regulated or managed, such as by chilling the temperature to a predetermined or selected temperature to accommodate aquatic live within the livewell **111**. The livewell **111** can comprise a container for housing water, or another liquid. In one embodiment, the livewell **111** can be capable of storing at least one fish or aquatic animal. In another embodiment, the livewell **111** can be coupled to the fishing vessel **102**. The livewell **111** can be coupled to the fishing vessel **102** in a variety of ways, including, without limitation, fixedly attached or detachably attached to the fishing vessel **102**. In yet another embodiment, the livewell **111** can be located on or adjacent to land, such as near a body of water or on a dock. However, it will be understood that the livewell **111** can be located at any other location on land. The livewell **11** can be adapted to store and keep aquatic life alive and healthy for various reasons, including, without limitation, to store or transport the aquatic life, and for other reasons. Such aquatic life can include freshwater fish, such as crayfish, minnows, freshwater bait fish and fish bait, caught fish and other aquatic life and the like, and saltwater fish, such as shrimp, lobster, saltwater bait fish and fish bait, caught fish and aquatic life, and the like.

[0036] The present invention relates to an apparatus and system, and a method for installing the same, for managing the temperature of a livewell or other container housing water. The modular apparatus **100** can generally comprise: (a) a self-contained fluid line **110**; (b) a compressor **112**; (c) an expansion valve **114**; (d) a first fluid line **120**; and (e) a condenser **130**. In one embodiment, the modular apparatus **100** can further comprise an evaporator **150**.

[0037] Referring to FIGS. **2-4**, the self-contained fluid line **110** may comprise the compressor **112** coupled to and/or in fluid communication with the expansion valve **114** and be adapted for circulating a desired medium (not shown), including, without limitation, a refrigerant, such as fluorocarbons, chlorofluorocarbons, various freon types, ammonia, sulfur dioxide, propane and other non-halogenated hydrocarbons, and the like. The self-contained line **110** may confine the desired medium in a flexible hose or hoses (not shown), or through the use of other flexible tubing, including copper tubing, flexible piping, and future discovered materials, whether presently known or later developed.

[0038] In one embodiment, the compressor **112** can be adapted for converting the desired medium of the self-contained fluid line **110**, including, without limitation, a refrigerant, from one state to another state. In one embodiment, the compressor **112** can be adapted for converting a refrigerant from a liquid state to a gaseous state, which can aid in the circulation of the refrigerant in the self-contained fluid line **110**.

[0039] The expansion valve **114** can be adapted for converting the desired medium of the self-contained fluid line **110**, including, without limitation, a refrigerant, from one state to another state. In one embodiment, the expansion valve **114** can be adapted for converting a refrigerant from a gaseous state to a liquid state by decreasing the pressure of the refrigerant or other fluid, which can aid in the circulation of the refrigerant in the self-contained fluid line **110**.

[0040] Therefore, in combination, the self-contained fluid line **110** can be adapted to convert a desired medium, such as a refrigerant, from liquid state to a gaseous state, via the compressor **112**, and back to a liquid state, via the expansion valve **114**. Such phase transitions being capable of aiding in the temperature management nature of the self-contained fluid line.

[0041] The first pump **122** may further connect to the first fluid line **120**. The first fluid line **120** may be in communication with the self-contained fluid line **110**. The self-contained fluid line **110** can be adapted to remove heat from a desired liquid flowing within or through the first fluid line **120**. The first pump **122** may be coupled to and/or in fluid communication with a first inlet port **124** and a first outlet port **126** of the apparatus **100**. The first pump **122** can be adapted for selectively displacing or moving a desired liquid between the livewell **111** and the modular apparatus **100**. In one embodiment, the first pump **122** can generally comprise a hydraulic pump. The first inlet port **124** and the first outlet port **126** may be in fluid communication with the livewell **111**, another water-storage container, and/or any additional livewells. In one embodiment, the first inlet port **124** and the first outlet port **126** are located on a side or panel of an enclosure surrounding the apparatus **100** (see FIG. **9**). The first fluid line **120** may confine the desired liquid in a flexible hose or hoses (not shown), or through the use of other flexible tubing, including copper tubing, PVC, other non-corrosion sea water resistant material of pipes, joints, and sections, whether presently known or later developed. It will be understood that the first fluid line **120**, and the components thereof, can be adapted for interchangeably using different desired liquids, including, without limitation, the interchangeable use of freshwater and saltwater. In one embodiment, to achieve such desired interchangeability, the first fluid line **120**, and the components thereof, can comprise high-grade stainless-steel components, or other suitable corrosion-resistant materials.

[0042] The condenser **130** can be adapted to remove heat from the desired medium of the self-contained fluid line **110**. While a condenser **130** is described herein, the condenser **130** may comprise heat exchanger related parts and materials. As shown in FIGS. **1-4**, the condenser **130** can be adapted for a variety of arrangements, including, without limitation, as a forced air coil unit **140**, a liquid-cooled unit **240**, or a cold plate heat sink unit (not shown). In one embodiment, when the desired medium contacts or enters the condenser **130**, heat may be exchanged or transferred from the medium to another medium of lower temperature that may be stationary or in motion. Accordingly, the medium that contacted or entered the condenser **130** may exit the condenser **130** in a different state, including, without limitation, at a lower temperature, than when it contacted or entered the condenser **130**. In another embodiment, the condenser **130** can be adapted for removing heat from the desired medium of the self-contained fluid line **110** while it is in circulation therein.

[0043] In one embodiment of the present invention, the condenser **130** can be a forced air coil unit, wherein the forced air coil unit may comprise a fan or other air or fluid circulator or agitator. In one embodiment of the present invention, the forced air coil unit can be adapted so that the fluid agitator is coupled with the self-contained fluid line **110**, and the self-contained fluid line **110** can be in communication with another desired medium (not shown) circulated or agitated by the fluid agitator. In another embodiment, the medium circulated or agitated by the fluid agitator can include air, including, without limitation, ambient air in a fishing vessel or other similar environments.

[0044] The modular apparatus **100** can further include flow sensors, which may be positioned at the first inlet port **124**, the first outlet port **126**, a second inlet port **244** and/or a second outlet port **246** to detect an amount of liquid moving through the first fluid line **120** and/or the second fluid line **242**. In another embodiment, the flow sensors are positioned within one of or all of the first fluid line **120** and/or the second fluid line **242**. The flow sensors can determine whether liquid is moving within one or all of the first fluid line **120** and/or the second fluid line **242**. In yet another embodiment, the flow sensors may be positioned at the entry port **104** and the exit port **108**.
[0045] As best depicted in FIGS. **2**A and **3**, in one embodiment of the present invention, the condenser **130** can be a liquid-cooled unit **240** comprising the second fluid line **242**. As shown in FIG. **2**, the second fluid line **242** may be in communication with the self-contained fluid line **110** and the second pump **260**. The second pump **260** may be coupled to and/or in fluid communication with a second inlet port **244** and a second outlet port **246** of the apparatus **100**. The second pump **260** can be adapted for selectively displacing or moving a desired liquid through the second fluid line **242**, which can aid in the circulation of the desired liquid therein. In one embodiment, the second pump **260** can generally comprise a hydraulic pump.
[0046] While FIG. **2**A illustrates a modular apparatus with four ports and two fluid lines, a two-port, single fluid line example is also envisioned, as shown in FIG. **2**B. The embodiment shown in FIG. **2**B can be adapted to chill on-land applications, such as a swimming pool. As shown, the modular apparatus **100** embodiment shown in FIG. **2**B can include the self-contained fluid line **110**, the compressor **112**, the expansion valve **114**, the first fluid line **120**, the first inlet port **124**, the first outlet port **126**, the condenser **130**, and the evaporator **150**. Also, the modular apparatus **100** of FIG. **2**B can connect to the first pump **122**. These components can function in the same or similar method as the modular apparatus shown in FIG. **2**A. However, as stated above, the embodiment shown in FIG. **2**B can omit a second fluid line and a second pump.
[0047] The embodiment shown in FIG. **2**B can further include a fluid agitator **142**. The condenser **130** can be adapted to remove heat from the desired medium of the self-contained fluid line **110**. As shown in FIGS. **1-4**, the condenser **130** can be adapted for a variety of arrangements, including, without limitation, as a forced air coil unit **140**, a liquid-cooled unit **240**, or a cold plate heat sink unit (not shown). In one embodiment, when the desired medium contacts or enters the condenser **130**, heat may be exchanged or transferred from the medium to another medium of lower temperature that may be stationary or in motion. Accordingly, the medium that contacted or entered the condenser **130** may exit the condenser **130** in a different state, including, without limitation, at a lower temperature, than when it contacted or entered the condenser **130**. In another embodiment, the condenser **130** can be adapted for removing heat from the desired medium of the self-contained fluid line **110** while it is in circulation therein.
[0048] As illustrated in FIG. **1**, in one embodiment of the present invention, the condenser **130** can be a forced air coil unit **140**, wherein the forced air coil unit **140** may comprise a fan or other air or fluid circulator or agitator **142**. As best illustrated in FIG. **1**, in one embodiment of the present invention, the forced air coil unit **140** can be adapted so that the fluid agitator **142** is coupled with the self-contained fluid line **110**, and the self-contained fluid line **110** can be in communication with another desired medium (not shown) circulated or agitated by the fluid agitator **142**. In another embodiment, the medium circulated or agitated by the fluid agitator **142** can include air, including, without limitation, ambient air in a fishing vessel or other similar environments.
[0049] The embodiment of FIG. **2**B removes heat through a vent or other device rather than a second fluid line. In addition, the embodiment of FIG. **2**B can connect to any fluid reservoir through the first fluid line **120** and regulate the temperate of a fluid within the fluid reservoir.
[0050] The modular apparatus of FIG. **2**B can run on 220 V AC electricity because the modular apparatus may be regulating the temperature of a very large amount of water, such as water in a swimming pool. However, in some embodiments, the modular apparatus is salt water friendly and not limited to fresh water or water having chlorine. The modular apparatus of FIG. **2**B can further

connect to, either wirelessly or through a wired connection, a temperature probe, which may be located in an incoming pipe from the swimming pool or other liquid reservoir. This incoming pipe can also include a filter system, which can also be powered by a 220 V AC power source. The result of the modular apparatus of FIG. **2**B can chill water at a lower cost, as 220 V AC may be less expensive than 110 V power.

[0051] As shown in FIG. **3**, in another embodiment of the present invention, the modular apparatus **100** can include the first pump **122** and the second pump **260**, and the modular apparatus **100** can house and include the first pump **122** and the second pump **260**.

[0052] According to exemplary embodiments depicted in FIGS. **2** and **3**, the second fluid line **242** may confine the desired liquid in a flexible hose or hoses (not shown), or through the use of other flexible tubing, including copper tubing, whether presently known or later developed. The desired liquid confined by the second fluid line **242** can include, without limitation, lake or ocean water at an environmental temperature. The water can enter the apparatus at the second inlet port **244** and travel into and through the apparatus **100** to the second outlet port **246** via the second fluid line **242** under the influence of the second pump **260**. As such, the water can pass through the condenser **130** to transfer or remove heat from the medium that is circulating in the self-contained fluid line **110**. In one embodiment, the second inlet port **244** and the second outlet port **246** be located on a side or panel (not shown) of an enclosure (not shown) surrounding the apparatus **100**, which may be the same side or panel as the first inlet port and the first outlet port. It will be understood that the second fluid line **242**, and the components thereof, can be adapted for interchangeably using different desired liquids, including, without limitation, the interchangeable use of freshwater and saltwater. In one embodiment, to achieve such desired interchangeability, the second fluid line **242**, and the components thereof, can comprise high-grade stainless-steel components, or other suitable corrosion-resistant materials.

[0053] In another embodiment, the condenser **130** can be a cold plate heat sink unit, also known as a heat exchanger, wherein the cold plate heat sink unit can generally comprise at least one highly conductive material, including, without limitation, aluminum or copper. In one embodiment, the conductive material of the cold plate heat sink unit is capable of withdrawing heat from a medium, including, without limitation, a refrigerant, and dissipating the heat into ambient air or environments.

[0054] The present invention may further comprise an evaporator **150**. In one embodiment, the evaporator **150** can be coupled to and/or in fluid communication with the self-contained fluid line **110** and the first fluid line **120**. In one embodiment, the evaporator **150** can comprise coils, cold plates, or other suitable components capable of achieving the desired results. The evaporator **150** can be adapted to remove heat from a desired medium of the first fluid line **120**, including, without limitation, stationary, moving, or circulating liquids. In one embodiment, when the desired medium contacts or enters the evaporator **150**, heat may be exchanged or transferred from the medium to another medium of lower temperature that may be stationary or in motion. Accordingly, the medium that contacted or entered the evaporator **150** may exit the evaporator in a different state, including, without limitation, at a lower temperature, than when it contacted or entered the evaporator **150**. In another embodiment, the evaporator **150** can be adapted for removing heat from the desired medium of the first fluid line **120** while it is in circulation in the first fluid line **120**. In yet another embodiment, the evaporator **150** can be adapted for exchanging or transferring heat to the desired medium of the self-contained fluid line **110**.

[0055] As shown in FIG. **4**, in another embodiment, the present invention may further comprise a first fluid line control valve **410** and/or a first filter **420**. In one embodiment, the first fluid line control valve **410** may be coupled to and/or in fluid communication with the first fluid line **120**, the first pump **122**, the first inlet port **124**, and the first outlet port **126** of the apparatus **100**. The first fluid line control valve **410** can be adapted for managing the flow of the liquid that is selectively displaced or moved by the first pump **122**. In one embodiment, the first fluid line control valve **410** can be adapted for regulating the flow rate of water that is displaced or moved by the first pump **122**.

[0056] In another embodiment, the first filter **420** may be coupled to and/or in fluid communication with the first fluid line **120**, the first pump **122**, the first inlet port **124**, and the first outlet port **126** of the apparatus **100**. The first filter **420** can be adapted for removing debris and contaminants from the desired liquid that is moving through the first fluid line **120**, which can aid in the circulation of the desired liquid and the heat transfer between the self-contained fluid line **110** and the first fluid line **120**. However, it will be understood that the first filter **420** can be coupled to and/or in fluid communication with other components of the modular apparatus **100** to remove debris and contaminants as required to achieve the purposes of the present invention.

[0057] In one embodiment, the present invention may further comprise a second fluid line control valve and/or a second filter. The second fluid line control valve may be coupled to and/or in fluid communication with the second fluid line **242**, the second inlet port **244**, the second outlet port **246**, and the second pump **260** of the liquid-cooled unit **240** of the apparatus **100**. The second fluid line control valve can be adapted for managing the flow of the liquid that is selectively displaced or moved by the second pump **260**. In one embodiment, the second fluid line control valve can be adapted for regulating the flow rate of water that is displaced or moved by the second pump **260**, **360**.

[0058] In another embodiment, the second filter may be coupled to and/or in fluid communication with the second fluid line **242**, the second inlet port **244**, the second outlet port **246**, and the second pump **260** of the apparatus **100**. The second filter can be adapted for removing debris and contaminants from the desired liquid that is moving through the second fluid line **242**, which can aid in the circulation of the desired liquid and the heat transfer between the self-contained fluid line **110** and the second fluid line **242**.

[0059] While a filter is shown specifically with regard to FIG. **4**, any of the embodiments shown in FIGS. **1-4** may include a filter. Additionally, it should be noted that because the modular apparatus **100** circulated filtered and chilled water through the livewell **111**, clean water regularly enters the livewell **111**, which can decrease ammonium buildup due to the existence of live fish in a small container. Additionally, fish and other aquatic animals may become agitated when confined to a small water volume, resulting in bodily reactions such as vomiting or other bodily functions. Circulating fresh and filtered water in the livewell **111** generated by the modular apparatus **100** also expels and removes the bodily reactions caused by agitated fish. If these contaminants were not removed, fish may become “green eyed” or die, resulting in a disqualification of the caught fish during a fishing competition. Water chilled to a proper temperature also prevents agitation because the cold-blooded fish are not forced to adjust bodily temperature dramatically. Thus, the modular apparatus demonstrates an improvement over prior art livewells.

[0060] Although the embodiments discussed herein may generally identify the exchange or transfer of heat from the livewell in a removal manner, it will be understood that the present invention, and the components thereof, can be adapted and arranged for purposes of exchanging or transferring heat to the livewell or other container housing water. In an exemplary embodiment, the evaporator **150** can be adapted for exchanging or transferring heat from the desired medium of the self-contained fluid line **110** to the first fluid line **120**. In a further exemplary embodiment, the condenser **130** can be adapted for exchanging or transferring heat to the self-contained fluid line **110**, including heat from a medium of ambient or higher temperature, whether as a forced air coil unit **140**, a liquid-cooled unit **240**, or a cold plate heat sink unit.

[0061] As best shown in FIGS. **5-10**, the present invention may further comprise a modular enclosure **500**. The enclosure may comprise a front panel **510**, a first side panel **520**, a second side panel **530**, a rear panel **540**, a top panel **550**, and a bottom panel **560**. In one embodiment, the front panel **510** may be coupled with the first side panel **520** and the second side panel **530**, such that the front panel **510** contacts the edges of the respective side panels **520**, **530**. At least one of the first side panel **520** and the second side panel **530** may extend generally perpendicular from the front panel **510**. The rear panel **540** may be coupled to the first side panel **520** and the second side panel **530** in a manner similar to the front panel **510**. In such an embodiment, the rear panel **540** may be generally opposite of the front panel **510**. In one embodiment, the rear panel **540** may be oriented generally parallel to the front panel **510**.

[0062] In another embodiment, the top panel **550** can be coupled with the front panel **510**, the first side panel **520**, the second side panel **530**, and the rear panel **540**, such that the top panel **550** contacts the edges of the respective elements. At least one of the front panel **510**, the first side panel **520**, the second side panel **530**, and the rear panel **540** may extend downwardly in a generally perpendicular manner from the top panel **550**. In yet another embodiment, the bottom panel **560** may be coupled with the front panel **510**, the first side panel **520**, the second side panel **530**, and the rear panel **540** in a manner similar to the top panel **550**. In such an embodiment, the bottom panel **560** may be generally opposite of the top panel **550**. In another embodiment, the top panel **550** may be oriented generally parallel to the bottom panel **560**.

[0063] In one embodiment, the front panel **510**, the first side panel **520**, the second side panel **530**, the rear panel **540**, the top panel **550**, and the bottom panel **560** may be coupled together by a plurality of fastening means. Such fastening means may include, without limitation, bolts, rivets, screws, pins, clamping members, glue and adhesive materials, any combination thereof, and any other suitable fastening means, whether presently known or later developed. In another embodiment, the front panel **510**, the first side panel **520**, the second side panel **530**, the rear panel **540**, the top panel **550**, and the bottom panel **560** may be coupled through welding, gluing, any combination thereof, and any suitable joining means, whether presently known or later developed.

[0064] As shown in FIGS. **5-10**, in one embodiment, the front panel **510**, the first side panel **520**, the second side panel **530**, the rear panel **540**, the top panel **550**, and the bottom panel **560** may each generally define a square shape, and in combination, the enclosure **500** may generally define a generally compact cube shape. However, it will be understood that the front panel **510**, the first side panel **520**, the second side panel **530**, the rear panel **540**, the top panel **550**, and the bottom panel **560** may each separately define any variety of suitable symmetric and non-symmetric geometric shapes, including varying and different shapes from one another, suitable for the purposes of the present invention. The dimensions of the front panel **510**, the first side panel **520**, the second side panel **530**, the rear panel **540**, the top panel **550**, and the bottom panel **560** can all be such that the modular enclosure **500** is capable of being located or placed in a variety of areas, including areas with limited available space for storing the enclosure **500**.

[0065] As best shown in FIG. **6**, the front panel **510** may define a door or access panel **600** and comprise a lock **610** for securing the access panel **600**. In one embodiment, the entire front panel **510** may comprise the access panel **600**, as shown in FIG. **6**, but it will be understood that the access panel **600** can consist of any lesser portion or portions of the front panel **510**. The access panel **600** can be hingedly attached to the enclosure **500** by a plurality of rotating mounting means **602**, such that the access panel **600** can be selectively pivoted and opened to access any contents within the enclosure **500**. In one embodiment, the mounting means **602** can be located on a lateral side of the front panel **510**, as shown in FIG. **6**, so that the access panel **600** can be selectively pivoted and opened by generally traversing a horizontal plane. However, it will be understood that the mounting means **602** can be located at the top or bottom edges of the front panel **510**, such that the door can be selectively pivoted and opened by generally traversing a vertical plane. In another embodiment, the lock **610** can be a keyed lock. However, it will be understood that the lock **610** can be any suitable locking means, including, without limitation, padlocks, deadbolts, lever handle locks, cam locks, dialed locks, any combination thereof, and any other suitable locking means, whether presently known or later developed.

[0066] As best shown in FIG. **7**, the first side panel **520** may comprise a first vent **700** and a first handle **710**. The first vent **700** can be adapted for permitting air flow into, out of, and generally through the enclosure **500** during the operation of the present invention. In one embodiment, where the condenser (not shown) is a forced air coil unit (not shown), the first vent **700** can introduce or remove the desired medium (not shown), such as ambient air, that is circulated or agitated by the fluid agitator (not shown) of the forced air coil unit. In one embodiment, the first vent **700** may comprise a plurality of first slots **702**. In another embodiment, the plurality of first slots **702** may be arranged in an array or columns and rows of first slots **702**. As shown in FIG. **7**, in one embodiment, the array of the plurality of first slots **702** may comprise three rows and twelve columns of first slots **702**. The first handle **710** can be adapted to permit a user to grasp and lift the enclosure **500** and its contents.

[0067] As best shown in FIG. **8**, the second side panel **530** may comprise a second vent **800** and a second handle **810**. The second vent **800** can be adapted for permitting air flow into, out of, and generally through the enclosure **500** during the operation of the present invention. In one embodiment, where the condenser (not shown) is a forced air coil unit (not shown), the second vent **800** can introduce or remove the desired medium (not shown), such as ambient air, that is circulated or agitated by the fluid agitator (not shown) of the forced air coil unit. In one embodiment, the second vent **800** may comprise a plurality of second slots **802**. In another embodiment, the plurality of second slots **802** may be arranged in an array or columns and rows of second slots **802**. As shown in FIG. **8**, in one embodiment, the array of the plurality of second slots **802** may comprise two rows of seven columns of second slots **802**. In combination with the first vent **710**, the second vent **810** can be capable of creating a circulating airflow through the enclosure **500** for purposes of aiding the condenser as it removes heat from the desired medium of the self-contained fluid line (not shown). The second handle **810** can be adapted to permit a user to grasp and lift the enclosure **500** and its contents. The second handle **810** can be used in conjunction with the first handle **710**.

[0068] As best shown in FIG. **9**, the rear panel **540** may comprise a first inlet port **124** and a first outlet port **126**. The first inlet port **124** and the first outlet port **126** may define operable openings in the rear panel **540** and, in one embodiment, comprise a threaded post for attaching, in a screwing manner, to external flexible hose or tubing (not shown) in fluid communication with a livewell (not shown). However, it will be understood that the first inlet port **124** and the first outlet port **126** may comprise any suitable arrangement for attaching to external hose or tubing. Such external hose or tubing can be capable of transferring a desired liquid from the livewell to the first inlet port **124** and from the first outlet port **126** back to the livewell. By transferring the desired liquid from the livewell, the desired liquid can be displaced or moved through the first fluid line (not shown) of the apparatus **100**, including, without limitation, through the use of the first pump (now shown). Also shown in FIG. **9**, the rear panel **540** may further comprise a second inlet port **244** and a second outlet port **246**. The second inlet port **244** and the second outlet port **246** may define operable openings in the rear panel **540** and, in one embodiment, comprise a threaded post for attaching, in a screwing manner, to external flexible hose or tubing (not shown) in fluid communication with a body of water (not shown), including, without limitation a creek, river, pond, lake, sea, ocean, and any manmade version of the same, whether presently known or later developed. However, it will be understood that the second inlet port **244** and the second outlet port **246** may comprise any suitable arrangement for attaching to external hose or tubing. Such external hose or tubing can be capable of transferring a desired liquid from the body of water to the second inlet port **244** and, in some embodiments, from the second outlet port **246** back to the body of water. By transferring the desired liquid from the body of water, the desired liquid can be displaced or moved through a second fluid line (not shown) of a liquid-cooled unit (not shown) of the apparatus **100**, including, without limitation, through the use of a second pump (now shown).

[0069] As best shown in FIG. **10**, the top panel **550** may comprise a user interface **1000** and at least one gauge **1010**. The user interface **1000** may further comprise an input module **1002** with input devices **1004** and an output module **1006** with output devices **1008**. The input module **1002** can be in communication with a controller (not shown). The controller can be configured to be instructed or directed by an individual or an operator via the input module **1002**, or in some similar manner. The input module **1002** can be configured to adjust, upon instruction or direction of an individual or an operator, one or more operational characteristics of the apparatus **100**, and any component thereof. The input module **1002** can comprise one or more devices, including, without limitation, the input devices **1004**. In one embodiment, the input devices **1004** can comprise programmable push buttons associated with predetermined or preset input values and characteristics. However, it will be understood that the input devices **1004** can comprise any means suitable for providing an input for the input module **1002**, including, without limitation, a keyboard, touchpad, touch screen, control, joystick, microphone with associated speech recognition software, and the like, whether presently known or later developed. The input module **1002** can be configured to facilitate remote operation at a location other than one that is immediately adjacent or proximate to the apparatus **100**. In another embodiment, the input module **1002** can be another device (e.g., mobile electronic device, smartphone (iOS or Android), laptop, fishing vessel control interface) in communication with controller via any wireless connection (Wi-Fi, Bluetooth, NFC, etc.).

[0070] The output module **1006** can be in communication with a controller. The output module **1006** can be configured to output or provide at least one signal, value, or characteristic upon instructions from the controller. In one embodiment, the at least one signal generated by the output module **1006** can be a digital or audible indication, such as a sound, to an operator that corresponds with desired and/or preset values, thresholds, calculations, algorithms, and/or the like for purposes of alerting the operator. In another embodiment, the at least one signal generated by the output module **1006** can be an electronic instruction to another device (e.g., mobile electronic device, smartphone (iOS or Android), laptop, fishing vessel control interface) in communication with the output module **1006** via any wireless connection (Wi-Fi, Bluetooth, NFC, etc.), such as a component of the apparatus **100** or an at least one processing unit (not shown). The at least one signal provided by the output module **1006** can be received by a device, computer, or processor, including, without limitation, the output devices **1008**, for purposes of manually or automatically adjusting one or more of the operational characteristics of the apparatus **100**. The output module **1006** can be configured to facilitate remote operation at a location other than one that is immediately adjacent or proximate the apparatus **100**.

[0071] In one embodiment, the output devices **1008** can comprise programmable display screens capable of displaying predetermined or preset output values and characteristics based on the instructions provided by a controller. However, it will be understood that the output devices **1008** can comprise any means suitable for providing an output via the output module **1006**, whether presently known or later developed. In one embodiment, the display screens of the output devices **1008** can comprise a device by which information can be instantaneously and visually presented to an operator of the apparatus **100** or to a remotely located monitor, manager, or operator of the apparatus **100**. For example, the display screen can provide visual feedback, including, without limitation, real-time indications of the operational characteristics of the apparatus **100**. The real-time indications can include, without limitation, temperature, entry temperature, exit temperature, temperature setting, voltage, pressure, volumetric flow rate, water oxygen levels, water pH levels, water ammonia levels, historical fault codes (e.g., compressor high voltage failure, compressor low voltage failure, unit high voltage, unit low voltage, outlet water temperature is to low, water outlet temperature probe failure, water inlet temperature probe failure, inlet water flow is to low, outlet water flow is to low, and water inlet and outlet temperature is to large), and other operational characteristics of the apparatus **100** Further, the displays may comprise a monitor or screen that is stationary in nature or that is mobile in nature. A mobile display can be a computer tablet, smart phone, personal data assistant (“PDA”), and/or the like. The output device **1008** can also exist nearby a fishing vessel's steering wheel or fishing seat so that a user can see the livewell temperature for any location or designated locations on the fishing vessel.

[0072] As shown in FIG. **10**, the apparatus **100** can comprise one user interface **1000** located on top panel **550** of the enclosure **500**. However, it will be understood that the apparatus can comprise any number of user interfaces **1000**, which can be located at any location on the enclosure **500** or the apparatus **100**, or can be located remotely at a location other than one that is immediately adjacent or proximate the apparatus **100**. Such remote use of the user interface **1000** can be achieved in close proximity to the apparatus **100**, such as on the fishing vessel (not shown) to which a livewell (not shown) can be attached, or at further distances, such as on the shore or at greater distances. The at least one gauge can be adapted for outputting certain predetermined or preset characteristics of the apparatus **100**, including, without limitation, certain pressure or pressures occurring throughout the apparatus **100**, and the components thereof, during operation.

[0073] In another embodiment, as shown in FIG. **11**, the present invention may further comprise one or more of the following: (a) a controller **1100**; and/or (b) an adaptable power interface **1110**. FIG. **11** depicts the enclosure **500** with the front panel **510**, and the access panel **600** thereof, in an open position, such that some of the internal components of the apparatus **100** are visible. The controller **1100** can be a device adapted to receive various inputs, generate desired outputs, and control various devices. In one embodiment, the controller **1100** can comprise a printed circuit board or multiple printed circuit boards coupled together. In another embodiment, the controller **1100** can comprise any electrical circuit or any combination of suitable electrical components and conductors, operably coupled together, whether presently known or later developed. In one embodiment, the controller **1100** can be adapted for use with wireless devices via Wi-Fi, Bluetooth, and the like. Although FIG. **11** depicts one controller **1100**, it will be understood that the apparatus **100**, according to the present invention, can contain any suitable number of controllers **1100**. Additionally, the controller **1100** can comprise a programmable processor or other advanced controller implementing software applications designed to chill water in the livewell as well as monitor and detect errors related to chilling the water in the livewell, which is described in further detail below.

[0074] The controller **1100** can be adapted, on-site and off-site via the internet, satellite, and other unknown carriers of communication of future technologies, to allow an individual or individuals to have the ability to control the various elements of the invention. In one embodiment, the controller **1100** can be in communication with an input module (not shown) and an output module (not shown). In another embodiment, the controller **1100** can be in communication with and generally control various components of the apparatus **100**, including, without limitation, the compressor **112**, the first pump **122**, the condenser **130**, the fluid agitator, the second pump **260**, **360**, the first fluid line control valve **410**, and/or the second fluid line control valve. In a preferred embodiment, the controller **1100** can adjust certain operational characteristics of the apparatus **100** and the compressor **112**, the first pump **122**, the condenser **130**, the fluid agitator **142**, the second pump **260**, **360**, the first fluid line control valve **410**, and/or the second fluid line control valve. In an exemplary embodiment, an individual or individuals can regulate the temperature of the livewell (not shown), either in degrees Fahrenheit or degrees Celsius, by selectively adjusting the speed of the first pump **122**, the fluid agitator **142**, and/or the second pump **260**, **360**. In another embodiment, the controller **1100** can automatically control the apparatus **100**, and the various components thereof, with little to no instruction from an operator.

[0075] In one embodiment, the controller **1100** can control the refrigeration components of the modular apparatus **100** (e.g. the compressor **112**, the first pump **122**, the condenser **130**, the fluid agitator, the second pump **260**, **360**, the first fluid line control valve **410**, and/or the second fluid line control valve) to cause the water in the livewell to have a desired temperature. The controller **1100** can interface with a temperature probe to determine the water temperature in the livewell, and cause the refrigeration components to turn off upon the livewell water reaching the desired temperature. The controller **1100** can instruct the refrigeration components to turn back on and again chill water entering the livewell in response to the temperature probe determining that the livewell water temperature is a predetermined number of degrees higher than the desired temperature (e.g., 2 Fahrenheit degrees higher). In this way, the livewell needs no ice, as in conventional livewell chilling methods. In some embodiments, the controller **1100** can also determine if the water in the livewell is too cold, and heat the water entering the livewell to cause the livewell water to reach the desired temperature.

[0076] Additionally, the controller **1100** can be programmed to turn off the modular apparatus in response to error codes or other detected statuses, such as a moving boat. This process is described in further detail below with regard to FIG. **19**.

[0077] The adaptable power interface **1110** can be adapted to control electricity or other electrical energy to energize at least one component of the invention upon the instruction or direction of an individual or an operator, and to otherwise facilitate the control of the apparatus **100** and the various components thereof. In one embodiment, the adaptable power interface **1110** can be coupled with or in communication with at least one component of the apparatus **100**, including, without limitation, the compressor **112**, the first pump **122**, the condenser **130**, the fluid agitator, the second pump **260**, **360**, the first fluid line control valve **410**, the second fluid line control valve, and/or the controller **1100**.

[0078] In a preferred embodiment, the apparatus **100** is capable of interchangeably operating with alternating current (“AC”) electric current, including 110-volt AC and 220-volt AC, and direct current (“DC”) electric current, including 12-volt DC, 24-volt DC, 36-volt DC, or 48-volt DC. The adaptable power interface **1110** can be adapted for use with a variety of voltages, including, without limitation, 110-volt AC, 220-volt AC, 12-volt DC, 24-volt DC, 36-volt DC, or 48-volt DC, from a variety of energy sources, including, without limitation, battery, deep-cycle battery, lithium material batteries and future discovery of advanced battery materials, commercial, solar-powered, wind-powered, water-powered, and the like. In an exemplary embodiment, the adaptable power interface **1110** can permit an operator to select between 24-volt DC and 36-volt DC electric current originating from a battery or batteries coupled to the troll motor of a fishing vessel to which the apparatus **100** is coupled. In another exemplary embodiment, the adaptable power interface **1100** can be used to allow AC electric current to control electricity or other electrical energy to energize at least one component of the invention, when the apparatus **100** is used in connection with a livewell located on or adjacent to land.

[0079] In one embodiment, an operator can alter the operating electric current of the apparatus **100**, from AC to DC or DC to AC, through the use of the adaptable power interface **1110**, depending on the available energy sources. In another embodiment, an operator can alter the operating voltage of the apparatus, including, without limitation, between 24-volt DC and 36-volt DC, through the use of the adaptable power interface **1110**, depending on the available energy sources. In yet another embodiment, an operator can alter the operating voltage of the apparatus through the use of certain connectors or ports provided on the controller (not shown) and/or a printed circuit board.

[0080] FIG. **12** shows an apparatus **100** for managing the temperature of a livewell in accordance with one embodiment of the present invention. The apparatus **100** depicted in FIG. **12** is shown without an enclosure. As best shown in FIG. **12**, the apparatus **100** can comprise a self-contained fluid line **110**, a compressor **112**, a first fluid line **120**, a condenser **130**, a forced air coil unit **140** comprising at least one fluid agitator **142**, an evaporator **150**, and a controller **1100**. The compressor **112** can be coupled to an energy source (not shown) by power cables **1200**. As depicted in FIG. **12**, the forced air coil unit **140** can comprise two fluid agitators **142** coupled with the self-contained fluid line **110**. As shown in FIG. **12**, in one embodiment, the controller **1100** can be a printed circuit board and be connected to the various components of the apparatus **100** by wiring **1210**. The apparatus **100** can further comprise insulation foam **1220** for insulating the various components thereof.

[0081] FIGS. **13-16** provide schematic representations of a system **1300** for managing the temperature of a livewell (not shown). As illustrated in FIG. **13**, the system **1300** can generally comprises an input module **1002**, an output module **1006**, a controller **1100**, and a temperature probe **1310**. The input module **1002** and the temperature probe **1310** can be in communication with the controller **1100**, and the controller **1100** can be in communication with the output module **1006**.

[0082] In one embodiment, the temperature probe **1310** can be a device adapted to sense or detect temperature, either in degrees Fahrenheit or degrees Celsius. The temperature probe **1310** can be coupled with a livewell, the self-contained fluid line (not shown), the first fluid line (not shown), and/or the condenser (not shown) for sensing or detecting the temperature thereof or any component thereto. In a preferred embodiment, the temperature probe **1310** can be adapted for sensing or detecting the temperature of the livewell. The temperature probe **1310** of the present invention can be a thermocouple, resistance temperature detector, or other suitable temperature-measuring device or sensor, whether presently known or later developed. The temperature probe **1310** can further be adapted for generating a temperature data value as a desired output. In a preferred embodiment, the temperature probe **1310** can be in communication with the controller **1100** and can communicate the temperature data value to the controller **1100**. The temperature data value can correspond with a temperature, in degrees Fahrenheit or degrees Celsius, of a specified environment or locale. It will be understood that the system **1300** of the present invention can contain any suitable number of temperature probes **1310**. The temperature probe **1310** can connect to the controller **1100** via any wired or wireless connection.

[0083] As best illustrated in FIG. **14**, in one embodiment of the present invention, the controller **1100** can generally comprise at least one processing unit **1400** configured to carry out instructions either hardwired as part of an application-specific integrated circuit or provided as code or software stored in a memory (not shown). In another embodiment, the at least one processing unit **1400** can be in communication with a memory.

[0084] As best illustrated in FIG. **15**, in another embodiment of the present invention, the system **1300** can further comprise a pressure probe **1500** that can be in communication with the controller **1100**. In one embodiment, the pressure probe **1500** can be a device adapted to sense or detect pressure, including relative pressure and absolute pressure. The pressure probe **1500** can be coupled with the self-contained fluid line (not shown), the first fluid line (not shown), the second fluid line (not shown), and/or the condenser (not shown) for sensing or detecting the pressure thereof or any component thereto. The pressure probe **1500** of the present invention can be a pressure sensor, static pressure probe, a multi-hole pressure probe, a pressure transducer sensor, or other suitable pressure-measuring device or sensor, whether presently known or later developed. The pressure probe **1500** can further be adapted for generating a pressure data value as a desired output. Additionally, the pressure probe **1500** can detect low or zero pressure to detect various error statuses. In a preferred embodiment, the pressure probe **1500** can be in communication with the controller **1100** and can communicate the pressure data value to the controller **1100**. The pressure data value can correspond with a pressure, in absolute or relative terms, of a specified environment or locale. It will be understood that the system **1300** of the present invention can contain any suitable number of pressure probes **1500**. The pressure probe **1500** can connect to the controller **1100** via any wired or wireless connection.

[0085] As best illustrated in FIG. **16**, in yet another embodiment of the present invention, the system **1300** can further comprise a memory **1600** that can be in communication with the controller **1100**. The controller **1100** can be configured to be instructed or directed by the memory **1600**, or in some similar manner. The memory **1600** can comprise a non-transient computer-readable medium or persistent storage device for storing data or other information for use by the system **1300** or generated by the input module **1002**, the temperature probe **1310**, at least one processing unit **1400**, and/or the pressure probe **1500**. In one embodiment, the memory **1600** can store the value or cumulative values of the temperature data values generated by the temperature probe **1310** and the pressure data values generated by the pressure probe **1500**. In another embodiment, the memory **1600** can store instructions in the form of code or software for the controller **1100** and/or at least one processing unit **1400**. The memory **1600** can be provided with the controller **1100** and/or at least one processing unit **1400**, or it can be provided remote from the controller **1100** and/or at least one processing unit **1400**. In yet another embodiment, the memory **1600** can comprise a plurality of modules or databases that contain various data values.

[0086] In a preferred embodiment, the output module **1006** and/or the controller **1100** can be in communication with and adapted for activating the first pump **122** and/or the second pump **260**, **360** for purposes of selectively displacing or moving, respectively, a desired liquid through the first fluid line **120** and/or a desired liquid through the second fluid line **242**, to aid in the circulation of the same. In one embodiment, by circulating the desired liquids in the respective fluid lines **120**, **242**, the system **1300** can be capable of exchanging or transferring heat from the livewell. In another embodiment, by circulating the desired liquids in the respective fluid lines **120**, **242**, the system **1300** can be capable of exchanging or transferring heat to the condenser **130**. In a preferred embodiment, when the value of the temperature data value generated by the temperature probe **1310** equals or exceeds a predetermined value, the controller **1100** can cause the first pump **122** and/or the second pump **260**, **360** to be activated via at least one signal. It will be understood, however, that the output module **1006** and/or the controller **1100** can be in communication with and adapted for activating any component of the apparatus **100** for purposes of exchanging or transferring heat from the livewell, including the compressor **112**, the condenser **130**, the fluid agitator, the first fluid line control valve **410**, and/or the second fluid line control valve.

[0087] Although the embodiments discussed herein may generally identify the use of a single apparatus **100** in connection with a single livewell, it will be understood that the present invention can include the use of any suitable number of apparatuses **100** in connection with any desired number or sizes of livewell or other containers housing water. In an exemplary embodiment, multiple apparatuses **100** can be used in conjunction to manage the temperature of a large livewell, including large livewells that may be located on land, or multiple livewells together. In an exemplary embodiment, the size of the livewell can be 1,000 gallons or larger, and several apparatuses **100** of similar size and arrangements can be provided for adequately managing the temperature of the same. However, it will be understood that there is no limit on the number of apparatuses **100** that can be used to manage a large livewell, and any number of apparatuses **100** according to the present invention can be used to manage the temperature of any large livewell regardless of the size. Further, the present invention may comprise multiple user interfaces **1000** used in connection with managing the temperature of a large livewell or multiple livewells. In one embodiment, the multiple user interfaces **1000** can comprise respective input modules **1002**, each configured to adjust one or more operational characteristics of the respective apparatus or apparatuses **100** used in the connection with the livewell or livewells, and output modules **1006**, each configured to output or provide at least one signal, value, or characteristic associated the respective livewell or livewells. Further yet, the present invention may comprise multiple temperature probes **1310** to aid in managing the temperature of a large livewell or multiple livewells. In one embodiment, the multiple temperature probes **1310** can be capable of detecting several temperatures, and generating associated temperature data values, at several locations of a large livewell or in multiple livewells, so that the temperature of the same can be properly managed by the respective apparatuses. In these ways, multiple modular apparatuses **100** can be adaptively used in conjunction to adequately manage the temperature of any livewell or any arrangement livewell, without demanding large amounts of space for the apparatuses **100** or the enclosures **500** of the apparatuses **100**.

[0088] Additionally, although the exemplary embodiments describe the modular apparatus **100** as interfacing with a fishing vessel, the modular apparatus can control fluid temperature on land or at sea. The modular apparatus need only connect to any fluid container to control and regulate temperature therein.

[0089] According to exemplary embodiments, a method or process of installing a system **1300** for managing the temperature of a livewell, of the type presented herein, can also be provided with the present invention. FIG. **17** is a diagram depicting an example method **1700** for installing the system **1300**. As indicated by block **1710**, existing livewell hose can be decoupled and removed from a livewell. In one embodiment, the livewell can be coupled to a fishing vessel, and the livewell hose may be in fluid communication with the body of water in which the fishing vessel is operated. Therefore, it may be necessary to remove the livewell hose connecting the livewell to the body of water. Such a process may require restricting the ability of water from the body of water from entering the fishing vessel. Block **1720** illustrates how, after the livewell hose has been decoupled, a self-contained fluid line and a first fluid line can be provided, and the first fluid line can be coupled with the livewell. In one embodiment, an enclosure can be provided for housing the self-contained fluid line and the first fluid line, and the first fluid line can be coupled with the livewell via a first inlet port and a first outlet port defining operable openings of the enclosure. In a preferred embodiment, the self-contained fluid line can be in communication with the first fluid line, including via an evaporator. As indicated by block **1730**, the enclosure can be attached, including fixedly or detachably, to an area adjacent the livewell. In one embodiment, this can include attaching the enclosure, whether fixedly or detachably, to a fishing vessel. In another embodiment, the enclosure **500** can be attached to a location adjacent a fishing vessel's **1800** stern or aft-most portion **1802**, including in dry storage compartments provided on the fishing vessel or under a floorboard of the fishing vessel, and adjacent to the livewell **1810**, as best shown in FIG. **18**. Where no such compartment is provided, one can be fabricated and made for the fishing vessel. Block **1740** illustrates how a user interface, comprising an input module and an output module, both of which can be in communication with a controller, can be provided and coupled with the enclosure. In one embodiment, the controller can be configured to be instructed or directed by an individual or an operator via the input module, or in some similar manner. In another embodiment, the output module can be configured to output or provide at least one signal, value, or characteristic upon instructions from the controller. In yet another embodiment, the controller can be in communication with and generally control, by adjusting certain operational characteristics of a compressor, a first pump, a condenser, a fluid agitator, a second pump, a first fluid line control valve, and/or a second fluid line control valve. The user interface can be provided at a location other than one that is immediately adjacent or proximate the apparatus **100**. As indicated by block **1750**, a temperature probe can be provided and adapted to be in communication with the controller. In one embodiment, the temperature probe can be placed in or adjacent to the livewell. In another embodiment, the temperature probe can generate a temperature data value. In a preferred embodiment, the temperature probe can communicate the temperature data value to the controller. Block **1760** illustrates how, an individual or an operator can provide or input a predetermined value into the user interface, such that when the value of the temperature data value equals or exceeds the predetermined value, the controller can adjust an operational characteristic of the compressor, the first pump, the condenser, the fluid agitator, the second pump, the first fluid line control valve, and/or the second fluid line control valve. In one embodiment, by adjusting an operational characteristic of the compressor, the first pump, the condenser, the fluid agitator, the second pump, the first fluid line control valve, and/or the second fluid line control valve, the controller can be capable of automatically exchanging or transferring heat from the livewell. In another embodiment, by adjusting an operational characteristic of the compressor, the first pump, the condenser, the fluid agitator, the second pump, the first fluid line control valve, and/or the second fluid line control valve, the controller can be capable of automatically exchanging or transferring heat to the condenser.

[0090] In one embodiment, the method or process of installing the system **1300** can further comprise the step of providing an adaptable power interface adapted to control electricity or other electrical energy to energize various components of the system upon the instruction or direction of an individual or an operator. In one embodiment, the adaptable power interface can be coupled with or in communication with the compressor, the first pump, the condenser, the second pump, the first fluid line control valve, the second fluid line control valve, and/or the controller. The adaptable power interface can be adapted for use with a variety of voltages from a variety of energy sources. In one embodiment, an operator can alter the operating electric current through the use of the adaptable power interface. In another embodiment, an operator can alter the operating voltage through the use of the adaptable power interface. In yet another embodiment, an operator can alter the operating voltage through the use of certain connectors or ports provided on the controller and/or a printed circuit board.

[0091] FIG. **19** illustrates a method **1900** for detecting statuses and controlling the modular apparatus based on detected statutes. As shown, the method **1900** can include the controller **11000** receiving a signal from one or more sensors (e.g. the temperature probe **1310**, the pressure probe **1500**, voltage sensor on controller **11100**) in step **1910**, and the controller **1100** further determining if the sensor reading corresponds to one of a plurality of error statuses in step **1920**. The plurality of statuses can include: low battery or no power coming from an AC power connection, a compressor high voltage failure, a compressor low voltage failure, a high voltage failure when the voltage from a power source is determined to be too high or does not match, a low voltage failure when the voltage from a power source is determined to be too low or does not match, water pressure at the outlet **126** is too low, temperature probe failure, pressure sensor failure, fluid pressure in the second fluid line is low-indicating the entry port **104** is not submerged within the body of water and the boat is moving. When the controller **1100** determines that one of the errors statuses exists, the controller **1100** turns off the modular apparatus **100** and, optionally, turns off any connected pumps in step **1930** (if not, the method returns to step **1910**). Subsequently, the controller **1100** determines whether all error statutes have cleared in step **1940**. If the controller **1100** determines that all the errors statuses have cleared, the controller **1100** turns the modular apparatus **100** back on and, optionally, turns on any connected pumps in step **1950** (if not, the method returns to step **1940**. After turning the modular apparatus **100** back on in step **1950**, the method returns to step **1910** and the controller **1100** returns to regulating the temperature within the livewell.

[0092] As an example of the method **1900**, the controller **1100** can determine that the boat is moving in step **1920** after receiving a signal from the pressure probe **1500** in the second fluid line **242** indicating that fluid pressure in the second fluid line **242** is low. The controller **1100** understands that a low pressure reading in the second fluid line **242** indicates that the entry port **104** on the fishing vessel **102** is not submerged within the body of water, and the pump **260** is not pulling in any water from the entry port **104** because the entry port **104** is elevated above the surface of the water and is not in fluid communication with the body of water.

[0093] It is important to note that the construction and arrangement of the elements of the inventive concepts and inventions as described in this application and as shown in the figures above is illustrative only. Although some embodiments of the present inventions have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible without materially departing from the novel teachings and advantages of the subject matter recited. All such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the preferred and other exemplary embodiments without departing from the spirit of the present inventions.

[0094] It is important to note that the apparatus of the present inventions can comprise conventional technology (e.g., as implemented in present configuration) or any other applicable technology (present or future) that has the capability to perform the functions and processes/operations indicated in the FIGURES. All such technology is considered to be within the scope of the present inventions and application.

Claims

1. A method for managing a temperature of a livewell on a boat comprising: regulating the temperature of the livewell using temperature regulation components of a modular apparatus by: detecting a temperature of water in the livewell using a temperature probe; controlling at least one of a compressor, a condenser, an evaporator, and an expansion valve to cause the temperature of the water in the livewell to reach a desired temperature using a controller, wherein the expansion valve is in fluid communication with the compressor; receiving a first signal from a sensor; determining whether the first signal corresponds to one or more error statuses; deactivating the temperature regulation components in response to determining that the first signal corresponds to the one or more error statuses, wherein the temperature regulation components include at least one of the compressor, the condenser, and the evaporator; receiving a second signal from the sensor; determining whether the second signal indicates that the one or more error statuses has cleared; and reactivating the temperature regulation components in response to determining that the second signal indicates that one or more error statutes have cleared.
2. The method of claim 1, wherein the sensor is a pressure sensor included in a fluid line, and wherein the one or more error statuses is a pressure low status indicating a low fluid pressure in the fluid line by determining that the first signal includes a pressure reading in the fluid line below a threshold, wherein the pressure reading is measured by the pressure sensor.
3. The method of claim 2, wherein the pressure low status indicates that an entry port connected to the fluid line is not submerged within a body of water outside the boat.
4. The method of claim 3, further comprising: determining that the boat associated with the livewell is moving to a new destination in response to determining the pressure low status.
5. The method of claim 4, further comprising: determining that the one or more error statuses have cleared in response to the second signal exceeding the threshold.
6. The method of claim 1, wherein the one or more error statuses includes a low battery, no power coming from an AC power connection, a compressor high voltage failure, a compressor low voltage failure, a high voltage failure when a voltage from a power source is determined to be too high, a low voltage failure when the voltage from a power source is determined to be too low, water pressure is too low, temperature probe failure, pressure sensor failure, or a combination thereof.
7. The method of claim 1, wherein the sensor is a pressure sensor within a fluid line, and wherein the one or more error statuses is a pressure low status indicating a low fluid pressure in the fluid line by determining that the first signal includes a pressure reading in the fluid line below a threshold.
8. The method of claim 1, further comprising: receiving the desired temperature via a wireless input module in communication with the controller.
9. The method of claim 1, further comprising: determining whether the water is flowing through a fluid line using a first flow sensor positioned at an entry port of the boat and a second flow sensor positioned at an exit port of the boat.
10. A method for managing a temperature of a livewell on a boat comprising: regulating the temperature of the livewell using temperature regulation components of a modular apparatus by: detecting the temperature of water in the livewell using a temperature probe in communication with a controller; controlling at least one of, a compressor a condenser, an evaporator, and an expansion valve to cause the temperature of the water in the livewell to reach a desired temperature using the controller, wherein the expansion valve is in fluid communication with the compressor; receiving a first signal from a sensor, wherein the sensor is a pressure sensor within a fluid line; determining whether the first signal corresponds to one or more error statuses, wherein the one or more error statuses is a pressure low status indicating a low fluid pressure in the fluid line when an entry port connected to the fluid line is not submerged within a body of water outside the boat; deactivating

the temperature regulation components in response to determining that the first signal corresponds to the one or more error statuses, wherein the temperature regulation components include at least one of the compressor, the condenser, and the evaporator; receiving a second signal from the sensor; determining whether the second signal indicates that the one or more error statuses has cleared; and reactivating the temperature regulation components in response to determining that the second signal indicates that one or more error statutes have cleared.

11. The method of claim 10, further comprising: removing ammonia from the livewell using a filter in fluid communication with a first inlet port and a first outlet port of the boat.

12. The method of claim 10, further comprising: determining that the one or more error statuses has cleared in response to the second signal exceeding a threshold.

13. The method of claim 10, wherein determining that the first signal corresponds to one or more error statuses includes determining that the first signal includes a pressure reading in the fluid line below a threshold, wherein the pressure reading is measured by the pressure sensor.

14. The method of claim 10, wherein the one or more error statuses includes a low battery, no power coming from an AC power connection, a compressor high voltage failure, a compressor low voltage failure, a high voltage failure when a voltage from a power source is determined to be too high, a low voltage failure when the voltage from a power source is determined to be too low, water pressure is too low, temperature probe failure, pressure sensor failure, or a combination thereof.

15. The method of claim 10, further comprising: receiving the desired temperature via a wireless input module in communication with the controller.

16. The method of claim 10, further comprising: determining whether the water is flowing through a fluid line using a first flow sensor positioned at an entry port of the boat and a second flow sensor positioned at an exit port of the boat.

17. A method for managing a temperature of a livewell on a boat comprising: regulating the temperature of the livewell using temperature regulation components of a modular apparatus by: detecting the temperature of water in the livewell using a temperature probe in communication with a controller; controlling at least one of a compressor, a condenser, an evaporator, and an expansion valve to cause the temperature of the water in the livewell to reach a desired temperature using the controller, wherein the expansion valve is in fluid communication with the compressor; receiving a first signal from a sensor, wherein the sensor is a pressure sensor within a fluid line; comparing the first signal to a threshold; determining that the first signal corresponds to one or more error statuses when the first signal is below the threshold, wherein the one or more error statuses is a pressure low status indicating a low fluid pressure in the fluid line when an entry port connected to the fluid line is not submerged within a body of water outside the boat; deactivating the temperature regulation components in response to determining that the first signal corresponds to the one or more error statuses, wherein the temperature regulation components include at least one of the compressor, the condenser, and the evaporator; receiving a second signal from the sensor; comparing the second signal to the threshold; determining that the one or more error statuses has cleared when the second signal exceeds the threshold; and reactivating the temperature regulation components in response to determining that the second signal indicates that one or more error statutes have cleared.

18. The method of claim 17, wherein determining that the first signal corresponds to one or more error statuses includes determining that the first signal includes a pressure reading in the fluid line below a threshold, wherein the pressure reading is measured by the pressure sensor.

19. The method of claim 17, wherein the one or more error statuses includes a low battery, no power coming from an AC power connection, a compressor high voltage failure, a compressor low voltage failure, a high voltage failure when a voltage from a power source is determined to be too high, a low voltage failure when the voltage from a power source is determined to be too low, water pressure is too low, temperature probe failure, pressure sensor failure, or a combination thereof.

20. The method of claim 17, further comprising: receiving the desired temperature via a wireless input module in communication with the controller.